Earth and Planetary Science Letters, 42 (1979) 121–126
© Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

[3]

NEGATIVE INCLINATION ANOMALIES FROM THE MEDICINE LAKE HIGHLAND LAVAS, NORTHERN CALIFORNIA

LAURIE BROWN

Department of Geology and Geography, University of Massachusetts, Amherst, MA 01003 (U.S.A.)

and

STANLEY A. MERTZMAN

Department of Geology, Franklin and Marshall College, Lancaster, PA 17604 (U.S.A.)

Received June 5, 1978
Revised version received October 6, 1978

Paleomagnetic directions have been determined for 21 flows of Quaternary age from the Medicine Lake Highlands in northern California. The mean direction for the area is $I = 49.3^\circ$ and $D = 3.8^\circ$ with an α_{95} of 5.4°. The corresponding mean VGP is located at 78.0°N and 42.4°E, a point which is offset 12° to the far, right-handed side of the rotation axis from the sampling site. The results here indicate only order of magnitude agreement with proposed models for offset poles. Together with other data from western United States there is an indication that large-scale regional field variations produce the observed inclination anomalies.

1. Introduction

To a first approximation, the time-averaged earth's magnetic field is an axial geocentric dipole [1,2]. Although this seems to hold true for most of the observed data, there have often been anomalous results which require alternative explanations. The most obvious inconsistency is the tendency for inclinations from Cenozoic units to be different from those predicted for the site latitude using the axial geocentric dipole. Possible explanations have included a tilt in the time-averaged dipole field [3]; a northward displaced axial dipole [4–6]; a combination of geocentric dipole and offset dipole [7,8]; and a general eccentric dipole calculated using spherical harmonic coefficients with observed data [9]. More recently Cox [10] has suggested that westward movement of the drifting part of the nondipole field could account for these inclination anomalies observed in Recent and Quaternary data. The VGPs determined from this study on 21 lava flows from the Medicine Lake Highlands show a larger than usual offset, and have corresponding inclinations too shallow for the axial dipole model by 12°.

2. Geology and petrology

The Medicine Lake volcano, located in extreme north-central California is part of the high-alumina plateau basalt petrologic province that lies parallel to and east of the High Cascades [11,12]. It is a large composite structure measuring approximately 40 km in basal diameter and rises 1000 m above the surronding Modoc Plateau. Lavas directly related to the volcano are dominantly andesitic with SiO_2 varying from 53 to 62% [13]. Occurring at the periphery of the Medicine Lake volcano and infrequently found intercalated with its andesites are flows of low-K, high-alumina olivine tholeiite and low-Si andesite. The Recent volcanic activity in the area is limited to these latter two groups; no Recent activity identi-

cal to the Medicine Lake andesites has been identified [13].

K-Ar age data for several samples analyzed paleomagnetically can be found in Table 1. Flow 2, the oldest flow investigated in this study, is 0.52 ± 0.21 m.y. old. This flow, together with flow 1, are from the Garner Mountain volcano [14]. Flow 12 (0.07 ± 0.04 m.y.), from one of the oldest andesite flows related to the Medicine Lake volcano, strongly suggests the 850–1000 m of lava extruded from the volcano was done so in a short span of geologic time, probably less than the last 100,000 years. Flow 13 (<0.15 m.y.) is from the youngest flow on the east rim of Mammoth Crater, a structure whose lavas lap onto the older Medicine Lake andesite flows and cover the western two-thirds of Lava Beds National Monument. The majority of the flows, including 5 through 11 and 14 through 17 are Recent in age, most being 1000–5000 years old [15]. The age data for these 21 flows, from K-Ar and stratigraphic relationships coupled with their normal polarity, indicate that all of them belong to the Brunhes Epoch.

The opaque mineralogy of the basalts consists of homogeneous titanomagnetite, often skeletal "quench texture" crystal outlines. Low-silica flows (48–50% SiO_2) normally have only groundmass opaques, excepting picotite inclusions in the olivine phenocrysts. In the andesite flows the opaque phenocrysts are small, <1 mm in diameter, and sparse, always <1% in terms of modal abundance. Titanomagnetite with variable amounts of exsolved ilmenite lamellae, together with discrete ilmenite, characterize these higher-silica flows. At no time has oxidation proceeded past class 3 in terms of the magnetite oxidation classification and class 2 in terms of the ilmenite oxidation classification of Wilson et al. [16]. Lastly, maghemite was not encountered in any of the flows we studied, suggesting low temperature post-solidification oxidation was not a significant factor affecting the paleomagnetic results.

TABLE 1

Analytical results of K-Ar dating (R. Walter, analyst) [a]

Sample	^{40}Ar [b]/g	^{40}Ar [b]/^{40}Ar × 100	K_2O (%)	K-Ar age (m.y.)
Flow 2 [c]	0.00368	1.87	0.48	0.52 ± 0.21
Flow 12 [d]	0.000906	0.69	0.85	0.07 ± 0.04
Flow 13 [e]	0.0000022	0.15	1.20	<0.15

[a] Constants used are: ^{40}K/total K = 1.19 × 10^{-4}; λ_β = 4.72 × 10^{-10} yr^{-1}; λ_e = 5.85 × 10^{-11} yr^{-1}.
[b] Radiogenic argon, 10^{-10} mol.
[c] Sample location: T.44N., R.1E., section 26, SE 1/4 SW 1/4, Bray quadrangle, California; for petrography and chemistry see Hughes and Mertzman [14] sample JH-74-46.
[d] See Mertzman [13] for sample description and exact location.
[e] Sample location: T.45N., R.4E., section 32, SW 1/4 SW 1/4, Medicine Lake quadrangle, California; 0.8% plagioclase phenocrysts <1 mm in diameter, 99.2% highly vesicular partially devitrified glass.

3. Field and laboratory procedures

Twenty-one flows, with a total of 169 cores, were sampled in the Medicine Lake Highlands area during June 1975. Identification and location of separate flows were done after three years of extensive field mapping and regional studies. Since there exists the possibility that lava blocks suffer reorientation after cooling [17], great care was needed in establishing sampling sites. Whenever possible, flow centers were sampled, a task made easier by old railroad cuts through the area. When good sections were not available, the most stable and well-defined part of the flow was used.

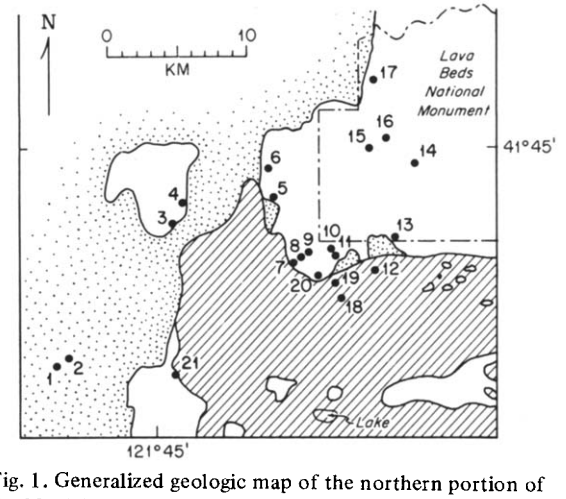


Fig. 1. Generalized geologic map of the northern portion of the Medicine Lake Highlands after Mertzman [13]. Stippled pattern = pre-Medicine Lake volcanics, ruled pattern = Medicine Lake related volcanics, plain pattern = post-Medicine Lake volcanics. Numbers are sampled flows and correspond to data in Table 1 and 2.

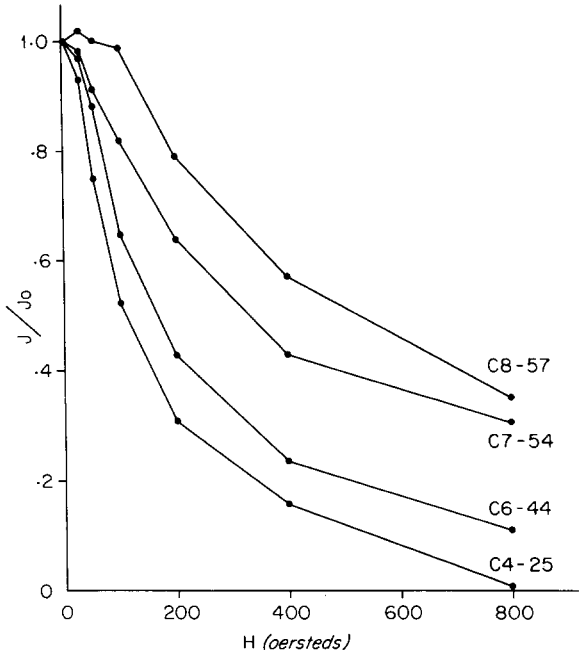


Fig. 2. Typical normalized AF demagnetization curves of Medicine Lake samples.

Six to eight cores were drilled in each flow, using a gasoline-powered rock drill. Cores were oriented in the field, and although Brunton compasses were used, back-siting was done to minimize errors in direction due to possible local magnetic anomalies. Locations of the sampled flows are shown in Fig. 1.

All the samples were measured on a PAR spinner magnetometer and demagnetized in a Schonstedt AC demagnetizing unit. Step-demagnetization studies, from 25 to 800 Oe, were done on a number of samples. In nearly every case the magnetization was exceedingly stable, and directions changed little even at 800 Oe, although intensity dropped markedly. From the step-demagnetization studies an optium level of 200 Oe was chosen, and all subsequent demagnetization work was done at this level. Typical demagnetization behaviors are shown in Fig. 2.

4. Paleomagnetic results

Paleomagnetic directions and poles are given in Table 2, along with associated Fisherian statistics.

TABLE 2

Flow-average directions of magnetization

Flow	Level	N	I	D	K	α_{95}	Lat.	Long.	Delm	Delp
1	H200	7	59.5	14.4	13	17.3	79.1	330.4	25.9	19.5
2	H200	8	66.4	2.7	81	6.2	82.5	251.6	10.2	8.4
3	H200	7	37.5	7.0	122	5.5	68.5	40.1	6.5	3.8
4	H200	7	36.8	355.7	21	13.5	68.5	69.2	15.8	9.2
5	H200	6	49.1	25.5	53	9.3	66.4	349.8	12.3	8.1
6	H200	6	46.8	4.2	69	8.1	75.9	43.2	10.5	6.8
7	H200	7	60.3	299.9	75	7.0	45.9	168.8	10.6	8.1
8	H200	6	49.9	10.8	209	4.6	76.0	16.6	6.2	4.1
9	H200	7	54.7	13.1	125	5.4	77.9	356.4	7.7	5.4
10	H200	6	47.9	2.5	36	11.3	77.1	48.7	14.7	9.6
11	H200	6	49.8	358.8	92	7.0	78.9	63.7	9.4	6.2
12	H200	6	67.0	339.6	60	8.7	73.7	184.8	14.5	12.0
13	H200	7	52.1	357.3	184	4.5	80.0	73.0	6.1	4.2
14	H200	7	49.2	1.5	250	3.8	78.3	52.0	5.1	3.4
15	H200	7	56.1	20.2	79	6.8	73.6	339.9	9.8	7.1
16	H200	8	58.7	5.3	105	5.4	85.3	357.0	8.1	6.0
17	H200	8	38.4	23.1	7	23.3	62.0	7.4	27.7	16.4
18	H200	7	34.4	1.9	37	10.1	67.2	53.7	11.6	6.6
19	H200	7	41.0	351.3	46	9.0	70.4	82.9	11.0	6.7
20	H200	8	50.2	10.6	23	11.9	76.3	16.4	16.0	10.7
21	H200	7	46.3	4.9	142	3.9	75.5	40.6	5.0	3.2

N = number of samples. I = magnetic inclination. D = magnetic declination. K = precision parameter. α_{95} = radius of cone of 95% confidence about resultant vector. Lat. and Long. = geographic coordinates of VGP. Delm and Delp = major and minor axes of oval of 95% confidence about the pole.

The majority of flows have directions which cluster well, with relatively small circles of confidence at the 95% level. Two flows show much wider scatter, with α_{95} values of 23.3° and 17.3°, and this divergence shows no sign of improving with increasing demagnetization levels. Such inconsistent flow directions have been linked to rotation of lava blocks after cooling below the Curie point [17], and general scatter from the aa and blocky characteristics of natural outcrops [18]. That does not seem to be the case here, as many natural outcrop sampling sites gave consistent results, and one of the questionable flows was sampled in a railroad cut where the supposedly stable center of the flow was accessible for drilling.

All flow directions are plotted in Fig. 3, with their circles of 95% confidence. The only flow separate from the rest is flow 7, which is also the youngest unit sampled. Alpha-95 circles for three of the flows include the axial dipole direction for northern California, and for two flows include the present earth's field direction. It is noted that one of the flows in each case has a very large scatter, and may be unreliable. It is also noted that the mean direction for the entire set is very similar to the direction obtained by Heinrichs [18] on Recent lava flows in the Three Sisters area of the Oregon Cascades.

Virtual geomagnetic poles are shown in Fig. 4.

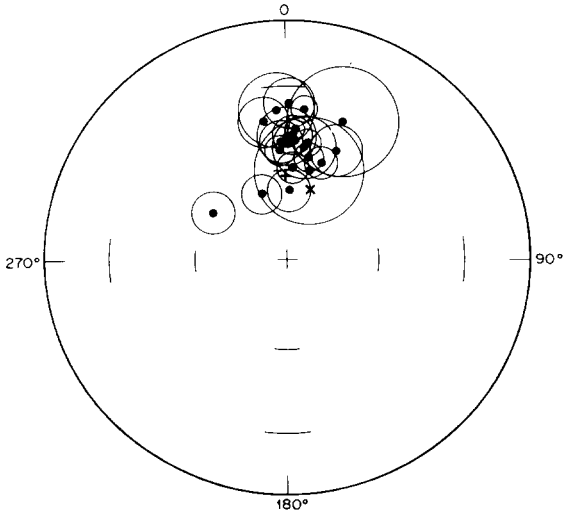


Fig. 3. Mean directions (inclination and declination) for all Medicine Lake flows with circles of 95% confidence. Plus sign is axial dipole field direction; cross is present geomagnetic field direction.

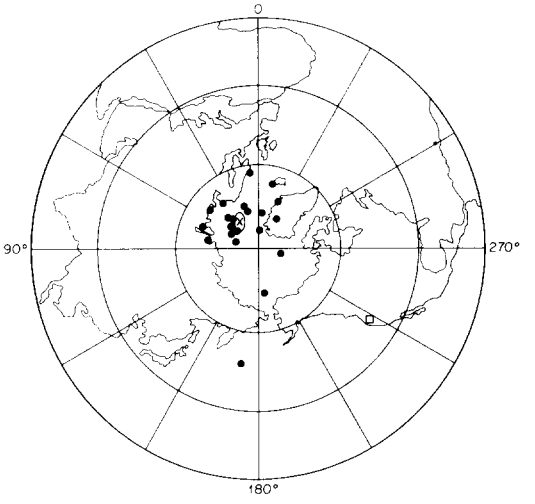


Fig. 4. Virtual geomagnetic poles for all Medicine Lake flows. Cross is mean VGP with 95% oval of confidence; square is sample site.

They cluster in one quadrant, far-sided and right-handed from the sample area. The two flows drastically different from the rest are flow 7 and flow 12. The latter is one of the oldest flows, so any attempt to relate the positions of the two to any field motion is impossible. The mean VGP is distinct from the rotation axis, and none of the ovals of confidence include the rotation pole. Again, data from the Cascades [18] give a very similar pole, as do data from the Kau series on Hawaii [19].

5. Discussion

The offset, far-sided pole position, noted by Wilson [4–6] is well manifested in this study. The average VGP at latitude 78.0° and longitude 42.4° is 12° beyond the rotation axis and right-handed with respect to the sampling site of northern California. This pole represents an average inclination of 49.3° and gives an inclination anomaly, ΔI, of −11.5° between the observed field and the axial geocentric dipole field. The anomaly is of the right sign but considerably larger than the −2.0° predicted by Cox's model [10] where the offset nature of poles is attributed to westward-drifting non-dipole components.

It is possible that one study may not represent an

TABLE 3

Inclination anomalies, Plio-Pleistocene studies from western United States

Location	Number of flows	I	D	Pole		ΔI	Reference
				lat.	long.		
Cascades, Oregon	9	49.6	2.2	76.1	50.3	−13.3	[18]
Tahoe area, California	23	45.3	22.7	67.3	356.8	−13.4	[20]
Sonoma, California	14	46.4	355.1	79.5	83.3	−11.3	[21]
Medicine Lake, California	21	49.3	3.8	78.0	42.4	−11.5	this study

I = magnetic field inclination. D = magnetic field declination. Pole, lat. and long. = virtual geomagnetic pole (VGP) position for study. ΔI = inclination anomaly.

average measure of the field's behavior over time. However, the results are similar to those from other studies on volcanic rocks of Pliocene-Pleistocene age from the western United States (Table 3). Directions from the Oregon Cascades [18], the Tahoe area [20] and Sonoma [21] all indicate inclination anomalies markedly larger than predicted by the models.

It is necessary to consider other possible causes for such a shallow inclination recorded in the Medicine Lake Highlands. There are no magnetic surveys over this area, although one exists to the west in the Mount Shasta area. Maximum anomalies observed are less than 2000 nT [22] and if indicative of local anomalies in the Medicine Lake area, are not significant enough to produce the reported directions. The possibility of erroneous directions being induced by the shape anisotropy of underlying flows has been suggested [23]. This possibility is difficult to evaluate, and although it can not be responsible for global patterns seen in inclinations [24], on a local scale it could have an appreciable effect. Since far-sided poles are reported for so many studies, it seems that a global model, either offset dipole or non-dipole components, is necessary. An attempt to distinguish between these two models using Icelandic lavas [25] supports the offset dipole model.

It is interesting to note that not only this study but also the other western United States studies (Table 3) all show right-handed offsets. This phenomenon has been reported previously [6,24], although it has recently been attributed to insufficient sampling [26]. Neither model provides any explanation for such an observation.

Our results suggest the possibility of a persistent, larger-than-usual standing non-dipole component of the field in the southern Cascades. The minimum field required to produce a 10° inclination anomaly for an average dipole field intensity of 50,000 nT would be 8700 nT. This is approximately half the value of the present non-dipole field. It certainly appears that the magnetic field is eccentric in the western United States and to a degree not explained by current models.

Acknowledgements

Field assistance in collecting samples was provided by P. Isaacson, S. Weaver and W. Hart. Reviews by P. Shive and M. Beck, comments and a preprint by C.G.A. Harrison, and editing by J. Reid, Jr. were most helpful. This work was funded in part by NSF grant EAR76-03763.

References

1 E. Irving, Paleomagnetism (Wiley, New York, N.Y., 1964).
2 N.D. Opdyke and K.W. Henry, A test of the dipole hypothesis, Earth Planet. Sci. Lett. 6 (1969) 139–151.
3 A. Cox and R.R. Doell, Long period variations of the geomagnetic field, Bull. Seismol. Soc. Am. 54 (1964) 2243–2270.
4 R.L. Wilson, Permanent aspects of the Earth's non-dipole magnetic field over Upper Tertiary times, Geophys. J. 19 (1970) 417–437.
5 R.L. Wilson, Dipole offset – the time average paleomagnetic field over the past 25 million years, Geophys. J. 22 (1971) 491–504.

6 R.L. Wilson, Paleomagnetic differences between normal and reversed field sources, and the problem of far-sided and right-handed pole positions, Geophys. J. 28 (1972) 295–304.
7 N.D. Watkins, Hemispherical contrasts in support for the offset dipole hypothesis during the Brunhes Epoch: the case for an unequal co-axial dipole pair as a possible geomagnetic field source, Geophys. J. 28 (1972) 193–212.
8 N.D. Watkins and A. Richardson, Analysis of Brunhes Epoch paleomagnetic data in terms of geocentric and offset axial dipole fields: long-term flattening of the dipole field, Geophys. J. 43 (1975) 501–516.
9 K.M. Creer, D.T. Georgi and W. Lowrie, On the representation of the Quaternary and Late Tertiary geomagnetic fields in terms of dipoles and quadrupoles, Geophys. J. 33 (1973) 323–345.
10 A. Cox, The frequency of geomagnetic reversals and the symmetry of the nondipole field, Rev. Geophys. Space Phys. 13 (1975) 35–51.
11 A.C. Waters, Basalt magma types and their tectonic associations: Pacific Northwest of the United States, Am. Geophys. Union, Geophys. Monogr. 6 (1962) 158–170.
12 M.W. Higgins, Petrology of Newberry Volcano, central Oregon, Geol. Soc. Am. Bull. 84 (1973) 455–488.
13 S.A. Mertzman, The petrology and geochemistry of the Medicine Lake volcano, California, Contrib. Mineral. Petrol. 62 (1977) 221–247.
14 J.M. Hughes and S.A. Mertzman, Geology and petrology of the Garner Mountain lavas, northeastern California, Northwest Geol. 5 (1976) 10–20.
15 S.A. Mertzman, Recent volcanism at Schonchin and Cinder Buttes, northern California, Contrib. Mineral. Petrol. 61 (1977) 231–243.
16 R.L. Wilson, S.E. Haggerty and N.D. Watkins, Variation of paleomagnetic stability and other parameters in a vertical traverse of a single Icelandic lava, Geophys. J. 16 (1968) 79–96.
17 D.E. James, Geology and rock magnetism of Cinder Cone lava flows, Lassen Volcanic National Park, California, Geol. Soc. Am. Bull. 77 (1966) 303–312.
18 D.F. Heinrichs, Paleomagnetic study of Recent Cascade lavas, Three Sisters, Oregon, J. Geophys. Res. 78 (1973) 5983–5992.
19 R.R. Doell, Paleomagnetism of the Kau volcanic series, Hawaii, J. Geophys. Res. 74 (1969) 4857–4868.
20 D.F. Heinrichs, Paleomagnetism of the Plio-Pleistocene Lousetown formation, Virginia City, Neveda, J. Geophys. Res. 72 (1967) 3277–3294.
21 E.A. Mankinen, Paleomagnetism and potassium-argon ages of the Sonoma volcanics, California, Geol. Soc. Am. Bull. 83 (1972) 2063–2072.
22 R.J. Blakely and R.L. Christiansen, The magnetization of Mount Shasta and importance for VGP determination from sea mounts, J. Geophys. Res. (in press).
23 R.S. Coe, Effects of shape anisotrophy on paleointensity determinations, Trans. Am. Geophys. Union 54 (1973) 1072.
24 R.L. Wilson and M.W. McElhinny, Investigation of the large scale paleomagnetic field over the past 25 million years. Eastward shift of the Icelandic spreading ridge, Geophys. J. 39 (1974) 570–586.
25 C.G.A. Harrison and N.D. Watkins, Comparison of the offset dipole and zonal non-dipole geomagnetic field models using Icelandic paleomagnetic data, J. Geophys. Res. (in press).
26 R.T. Merrill and M.W. McElhinny, Anomalies in the time-averaged paleomagnetic field and their implications for the lower mantle, Rev. Geophys. Space Phys. 15 (1977) 309–323.